

Wind Tunnel Turbulence Intensity Characterization

Part I: Shakedown Tests & Data Filtration	1
Part II: Part A Results – TI vs. Wind Tunnel Fan Speed	7
Part III: Part B Results – TI vs. Z-Position Data Collection	14

The purpose of this experiment was to determine the turbulence intensity found within air tunnel flow as a function of both fan speed and how the profile changes over the cross-section of the testing chamber. To collect results, three separate tests were completed to collect data accurately. Shakedown testing was completed to gain an error profile of measurements, taking into consideration sensor noise and drift, as well as confirming the start-up time of the fan to confirm accurate timing of data collection. Samples at separate fan speeds with controlled sample positioning were taken as well as samples at the same fan speed with modified locations in the testing cross-section.

Part I: Shakedown Tests & Data Filtration

Sampling Rate Selection

Before beginning the discussion and results of shakedown tests, it is important to note the sampling rate that was selected to collect data to measure turbulence intensity and correct model the turbulent behavior of air within the wind tunnel. It is known that low frequencies will be dominant in the response of pressure data from the sensor due to separation at the diffuser. Thus, a large enough frequency should be measured such that behavior of the fluid can be measured accurately and not sufficiently large such that significantly less data would characterize the flow effectively. A frequency rate of 1000Hz was selected. It should be known that due to Nyquist's criterion, the maximum frequency response that can be recorded is half the sample rate (500Hz). A 1000Hz sample rate is justified since turbulent kinetic energy occurs much below the 500Hz frequency mark, where nature of the fluid can be correctly analysed.

Shakedown Test 1 – Baseline Sensor Noise & Sensor Drift

The first shakedown test completed was used to measure baseline sensor to determine precision error of measurements. Additionally, the shakedown test was also utilized to determine bias uncertainty corresponding to sensor drift. As to be explained in greater detail within the second shakedown test, measurements from the baseline sensor noise test were used to better track sensor drift independently from drift induced by fan operation. To validate the results of the baseline noise and sensor drift, a FFT of test data as well as a wavelet spectrum of sensor data is evaluated to determine any deviations in expected results. As to be further justified in later sections within Part I, a low-pass filter (Butterworth) at 150 Hz was applied to all recorded pressure measurements. This was applied to reduce magnitude spikes found at the 200-400 range unrelated to flow characteristics as well as the natural frequency of the pitot tube and better isolate turbulent nature of the fluid.

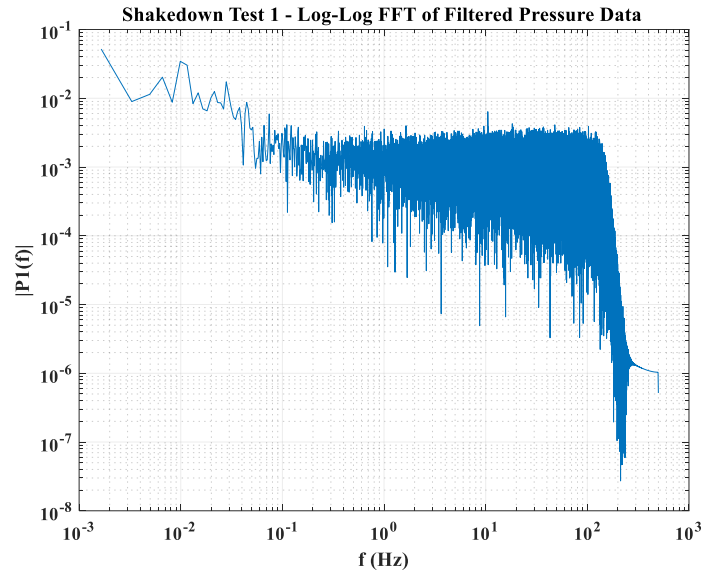


Figure 1: Shakedown Test 1 FFT Results

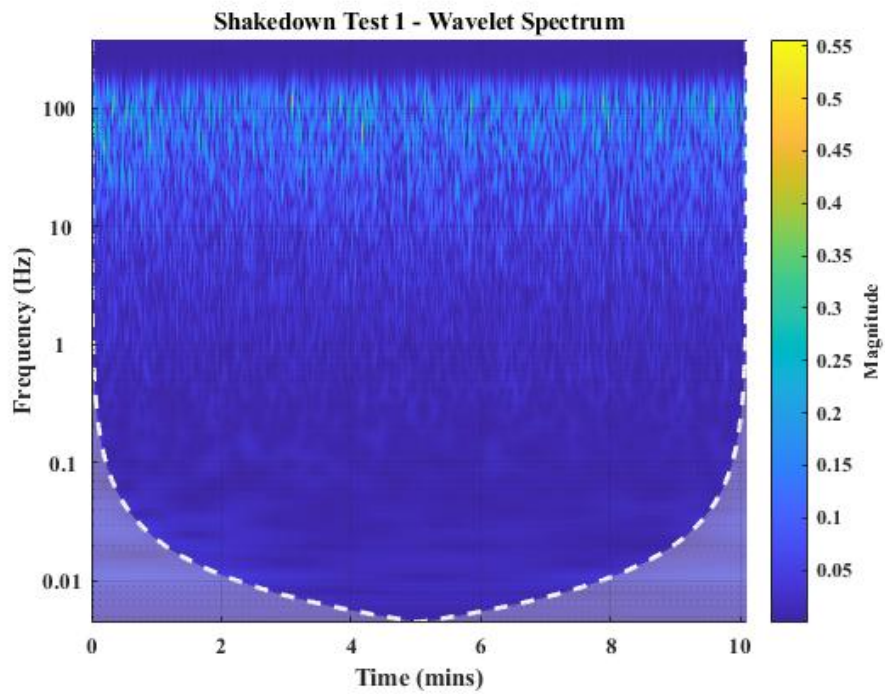


Figure 2: Shakedown Test 1 Wavelet Spectrum Results

Based on results from wavelet spectrum analysis and the FFT plot, no major spikes in magnitude are found through the duration of testing (10 minutes total).

Error Propagation Calculations

To calculate the overall uncertainty of velocity readings within the wind tunnel, the overall precision and bias uncertainty must be combined and converted from pressure to velocity. The precision uncertainty of pressure measurements due to the baseline noise test is calculated by finding the 95% confidence interval (Equation (1)).

$$U_{precision} = \pm \frac{z\alpha}{2}\sigma_x \quad (1)$$

$$U_{precision} = \pm z_{0.025}\sigma = \pm(1.96)(0.4894 \text{ Pa}) = \pm 0.96 \text{ Pa}$$

The total drift uncertainty was calculated using the difference in sample means of the first and last 100 sample points in data. Note that it is assumed that the difference in these means accurately reflect the bias of the sensor over the ten-minute testing period, and that the fan had fully reached steady state before data acquisition began. The precision uncertainty as found through analysis makes mathematical sense given the sensor range of $\pm 1 \text{ in H}_2\text{O}$ ($\sim 250 \text{ Pa}$).

$$\delta U_{\text{drift bias}} \approx \pm |\bar{x}_{\text{final}} - \bar{x}_{\text{initial}}| \quad (2)$$

$$\delta U_{\text{drift bias}} \approx \pm |\bar{x}_{\text{final}} - \bar{x}_{\text{initial}}| = \pm |0.1767 - 0.1878| = \pm 0.0111 \text{ Pa}$$

As expected, the bias uncertainty through sensor drift is minimal and produces nearly negligible effects on error propagation. Next, we find the drift bias because of linearity/hysteresis error as found in the 1 INCH-Dx-4V-MINI data sheet. Then, we can compute the overall uncertainty combining sources of both precision and bias error.

$$\delta U_{\text{lin,hyst}} = \pm \frac{0.25}{100} \text{FS}_{\text{sensor}} = \pm 0.0025(1 - (-1)) \pm 0.005 \text{ in H}_2\text{O} \cong \pm 1.24 \text{ Pa}$$

$$\delta U_{\text{tot}} = \sqrt{\sum \delta U_n^2} \quad (3)$$

$$\delta U_{\text{tot}} = \sqrt{\delta U_{\text{precision}}^2 + \delta U_{\text{drift bias}}^2 + \delta U_{\text{lin,hyst}}^2} = \sqrt{0.96^2 + 0.0111^2 + 1.24^2} \cong \pm 1.57 \text{ Pa}$$

Now, we must convert the uncertainty from pressure to meters per second to determine error plots for velocity data for varying wind speeds, separate z-locations in the test section, and for velocity intensity plots.

$$\delta V = \frac{dV}{d\Delta P} \delta U_{\text{tot}} \quad (4)$$

$$V = \sqrt{\frac{2\Delta P}{\rho}} \quad (5)$$

$$\delta V = \frac{dV}{d\Delta P} \delta U_{\text{tot}} = \frac{d}{d\Delta P} \left(\sqrt{\frac{2\Delta P}{\rho}} \right) \delta U_{\text{tot}} = \frac{\delta U_{\text{tot}}}{\rho V} = \pm \frac{(1.57)}{(1.225)V} = \pm \frac{1.22}{V}$$

It is important to note that the uncertainty from velocity readings is inversely proportional to velocity readings. This means that the lower the velocity, the higher the uncertainty. The

velocity uncertainty will be applied as a constant for any velocity data point used for tests within the lab for consistency.

Shakedown Test 2 – Drift Uncertainty

The second shakedown test performed through data collection of sample pressure data with the tunnel fan in operation. Although this test was initially planned to determine the total bias (drift) uncertainty, it was determined that extracting data while the fan is turned off would produce a more accurate depiction of drift due to the sensor itself rather than drift affected by wind tunnel operation. By operating the fan and inducing wind into the tunnel, affects due to the turbulent profile of the working fluid could cause the baseline drift to artificially increase. Thus, the drift uncertainty was determined utilizing data from shakedown test one, where the wind tunnel fan was not in operation. Testing was still completed for the shakedown tests, where log-log plots and the wavelet spectrum evidently display an increase in the magnitude of response considerably at low frequencies. Determining a mean at two separate points in data would cause the estimate to become significantly less accurate due to increased variance in sample data. The results of testing are shown in Figures (3) and (4). As expected, based on results, turbulent characteristics of flow become much more present in the wavelet spectrum at lower frequencies. 50Hz, the VFD setting tested for testing, is close to the maximum operating wind speed VFD setting (60Hz), so dominating frequencies on Figure (4) are to be expected.

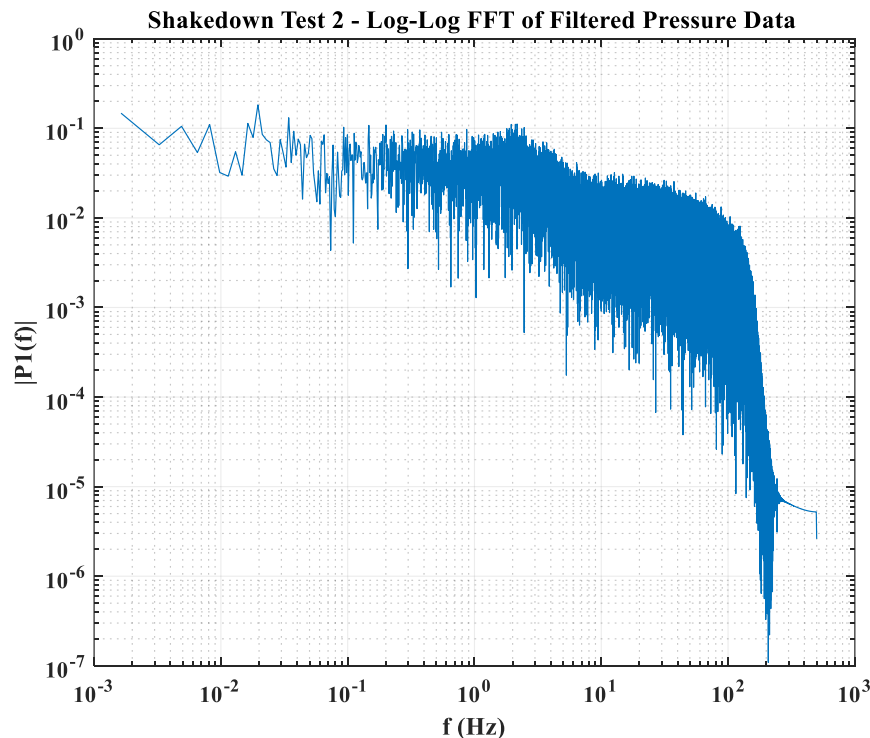


Figure 3: Shakedown Test 2 FFT Results

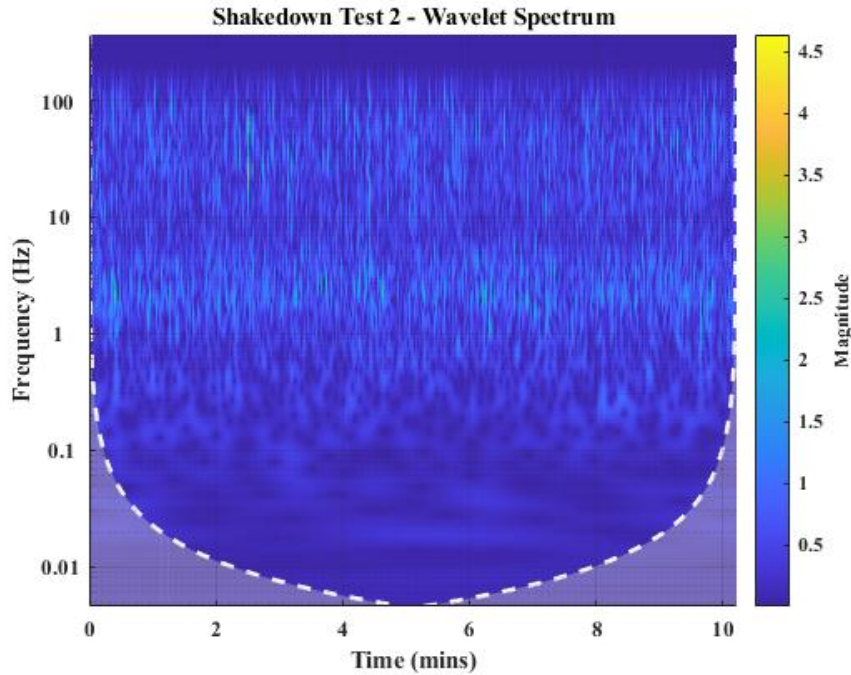


Figure 4: Shakedown Test 3 FFT Results

Shakedown Test 3

The final shakedown test completed was to determine the start-up time of the wind tunnel to reach a steady state operating speed, specifically to assist in noting if time-domain results from actual tests are collected while the wind tunnel has not fully reached steady state (constant wind speed). To visualize this data, a time-domain response forms the sensor output is displayed in units of pascals as shown in Figure (5).

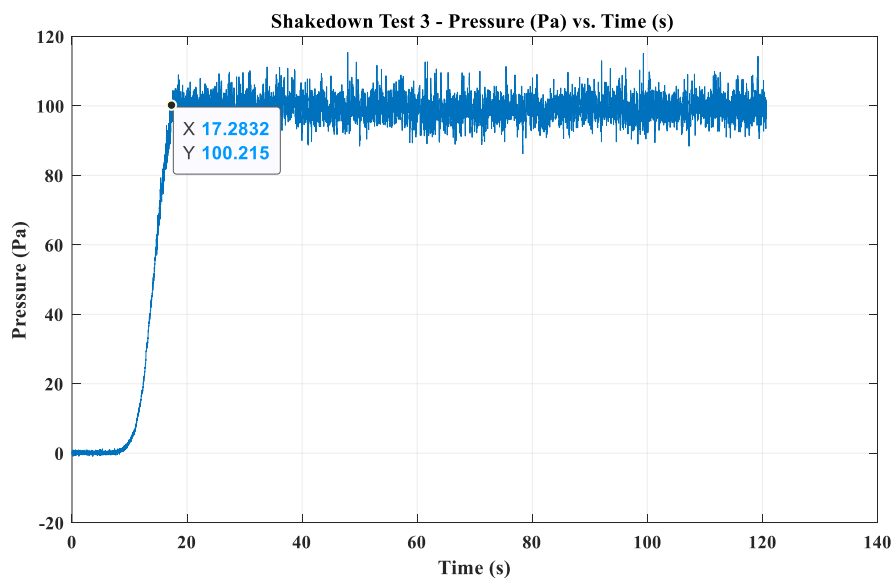


Figure 5: Shakedown Test 3 Time-Domain Results (Ramp-Up Time)

The wind-tunnel start-up time, or the time it took for the tunnel to reach steady state, was approximately 17 seconds. Testing was completed at 60Hz, the maximum fan operating speed, to the longest start-up time possible in testing. Although this shakedown test is not used for any statistical calculations, it is important to note that any output within the 0-17s second range as observed from other tests is not characteristic of the actual flow of the working fluid.

Filtration Techniques

To collect sufficient data, along with selecting an effective sample rate, filtration must be completed to filter out response frequencies that are not characteristic of the flow of the working fluid in the wind tunnel. To select an appropriate filter, data from shakedown Test 2 (VFD settings 50Hz) was placed in a FFT and wavelet spectrum to determine unwanted magnitude peaks. Based on Figure (6) below, peaks are recorded at frequencies >100Hz (243.35 & 377Hz), which are not characteristic of flow. These may be due to physical disturbances at the probe, including separation formed on the probe surface, vortex-induced vibrations, and the natural frequency of the probe. To determine if any frequency spikes were maintained between different tests, a 20Hz test from Part I was analysed pre-filtration on an FFT (Figure(7)), which captured the same frequency spike at 377Hz. It can be concluded that this frequency spike corresponds to the natural frequency of the probe. To remove the effects due to these frequencies, a Butterworth low-pass filter was applied with a cutoff frequency of 150Hz. As displayed in Part I and II results, dynamic behaviors of the fluid are clearly visible at frequencies well under 100Hz. Thus, a 150Hz filter conservatively preserves all frequencies that contribute to turbulent nature of the fluid. The effects of post-filtration are displayed clearly in Figures (1) & (3).

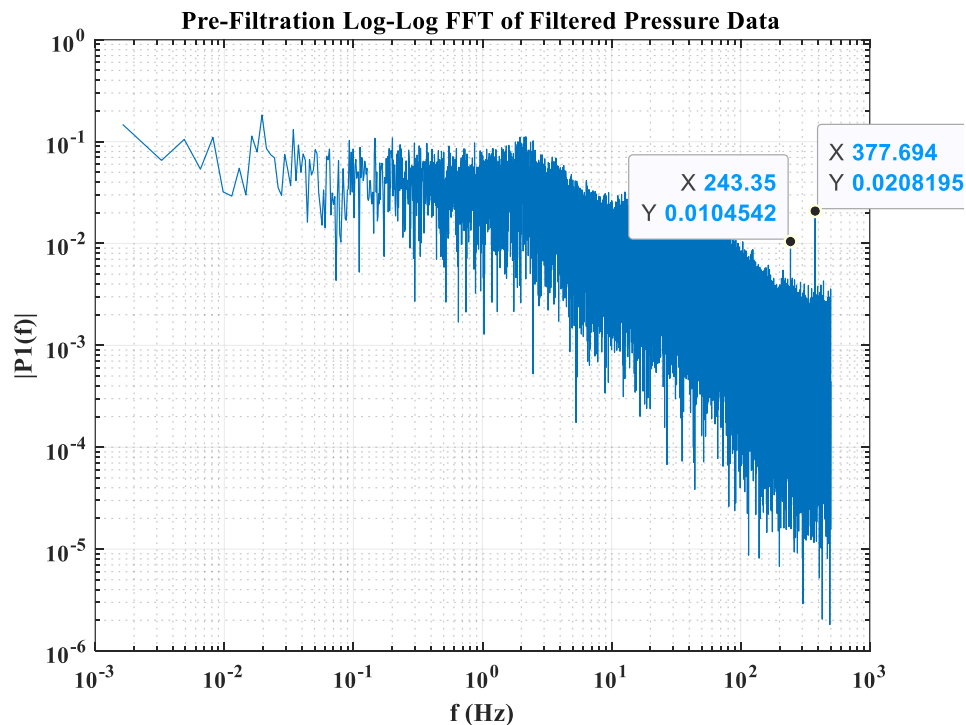


Figure 6: Pre-Filtration FFT (Shakedown Test 2)

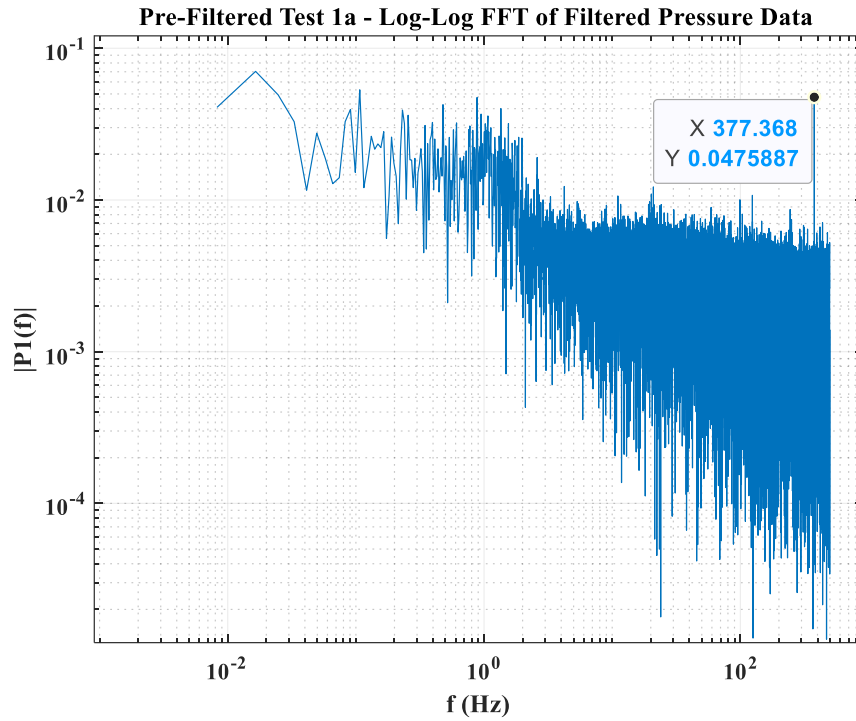


Figure 7: Post-Filtration FFT (Shakedown Test 2)

Part II: Part A Results – TI vs. Wind Tunnel Fan Speed

To convert pressure readings to velocity in all test cases, Equation (5) is utilized .

Test 1a - 20 Hz Free Stream Flow Results

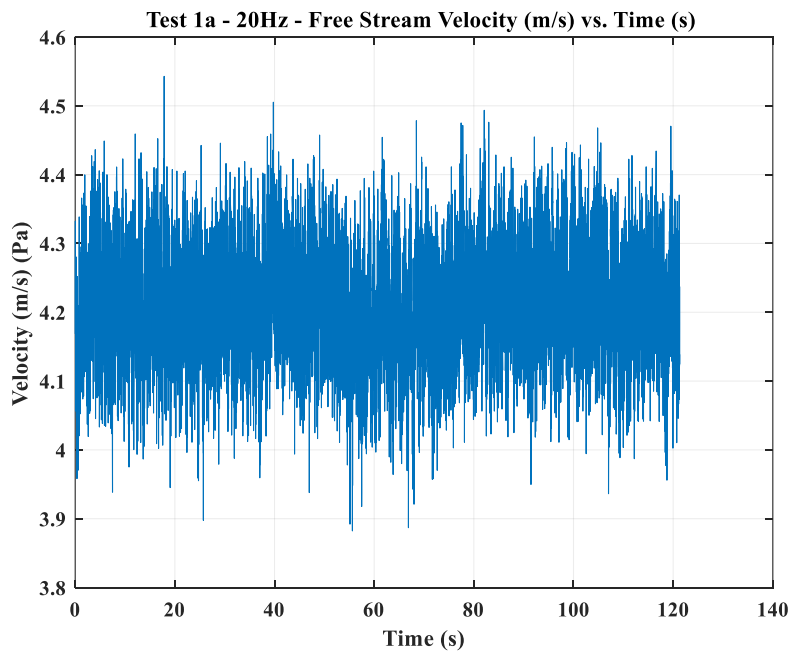


Figure 8: Test 1a (20Hz) Velocity vs. Time Response

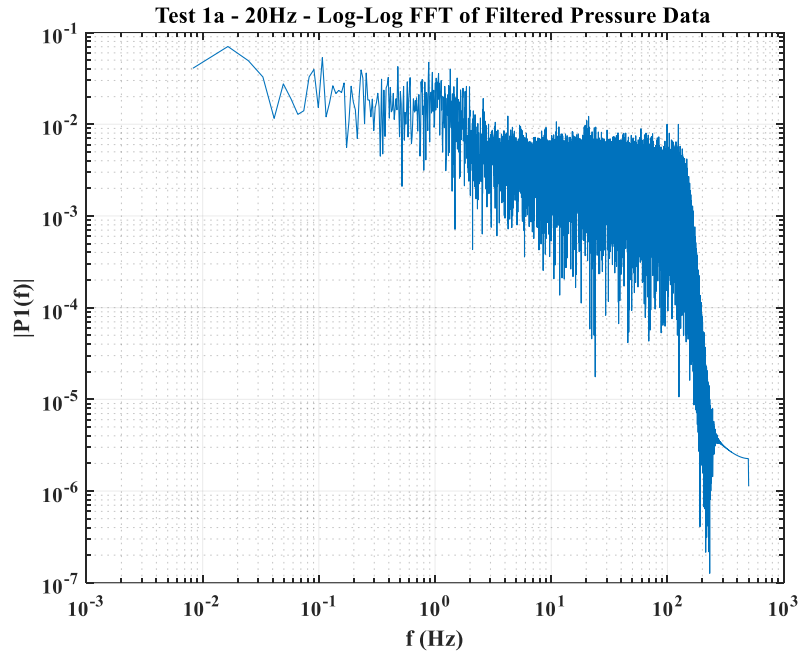


Figure 9: Test 1a (20Hz) FFT Plot

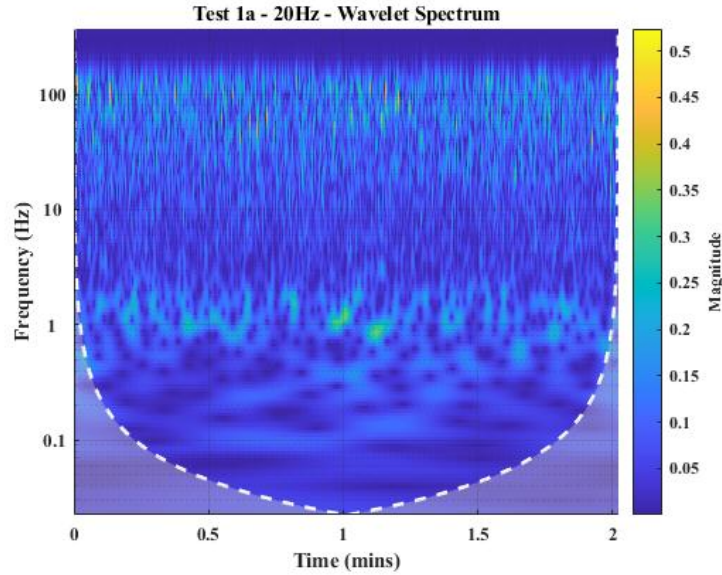


Figure 10: Test 1a (20Hz) Wavelet Spectrum

As displayed in the velocity vs. time, FFT, and wavelet plots above, the mean velocity ranges around 4.2m/s, which is approximately 1.3m/s then the expected velocity of the wind tunnel at the specified VFD controller setting. Additionally, we see high magnitude peaks at the central testing point (1 minute) around the 1Hz range. This decrease in air speed contributes greatly to an artificial increase in the velocity (and resultingly, turbulence intensity) at this speed. The low VFD setting applied in this test is the lower limit setting available. This could indicate the test catches non-linear behavior of the fan, reducing efficiently sufficiently.

Test 1b - 40 Hz Free Stream Flow Results

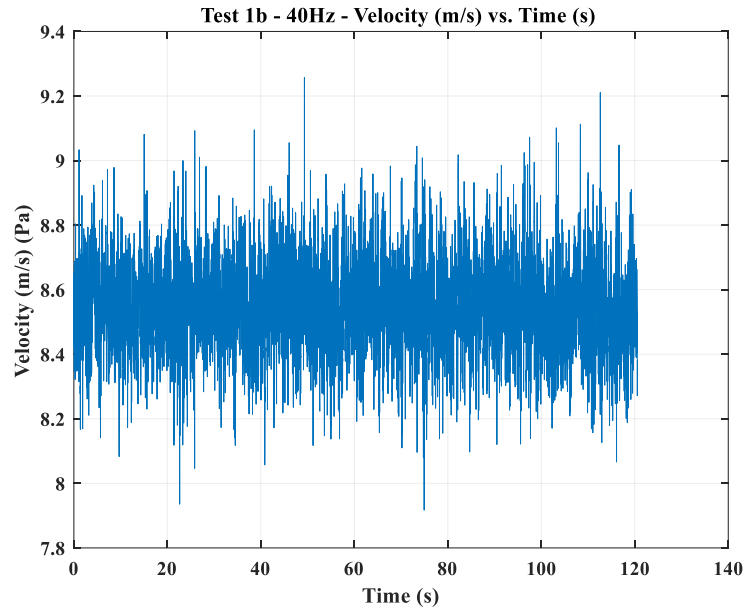


Figure 11: Test 1b (40Hz) Velocity vs. Time Response

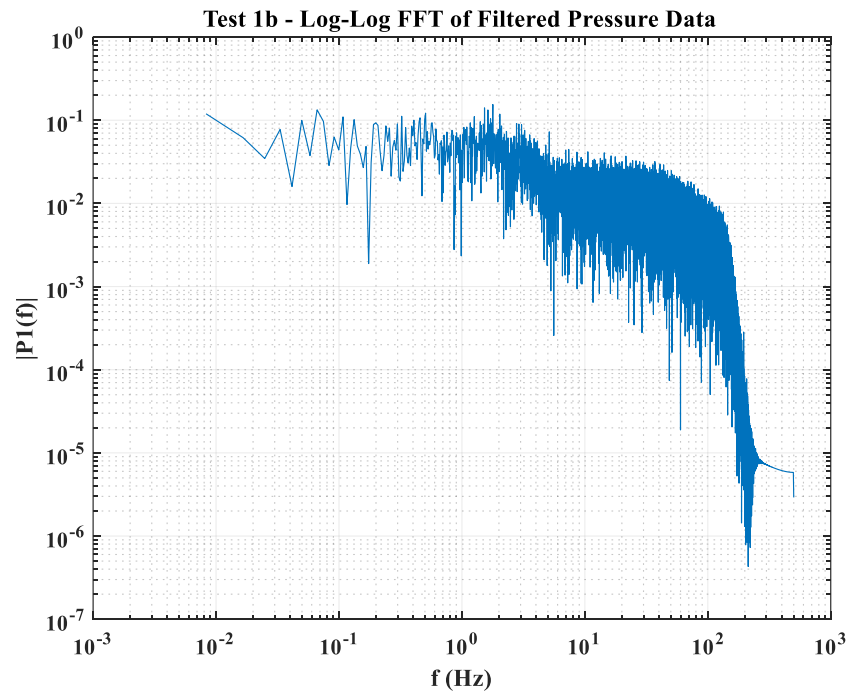


Figure 12: Test 1b (40Hz) FFT Plot

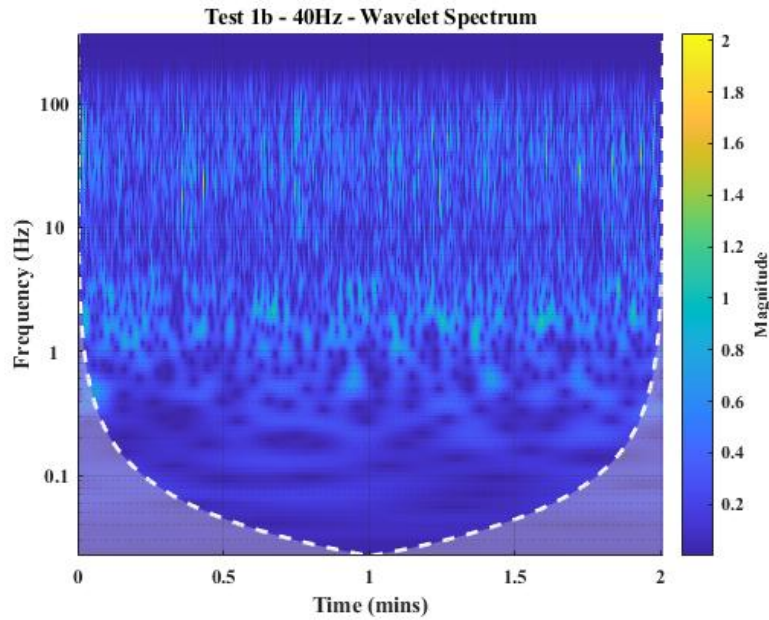


Figure 13: Test 1b (40Hz) Wavelet Spectrum

The 40Hz fan tests, compared to tests at 20Hz, produce significantly closer wind speeds to expected values. At a mean velocity of 8.54m/s, compared to an expected velocity of 8.5m/s, data collected at this fan speed is very accurate. We observe that the variance of sample data has increased, which is to be better visualized when plots are compared together. Additionally, it is useful to note that there are no outlier frequency spikes through the wavelet spectrum during test.

Test 1c - 60 Hz Free Stream Flow Results

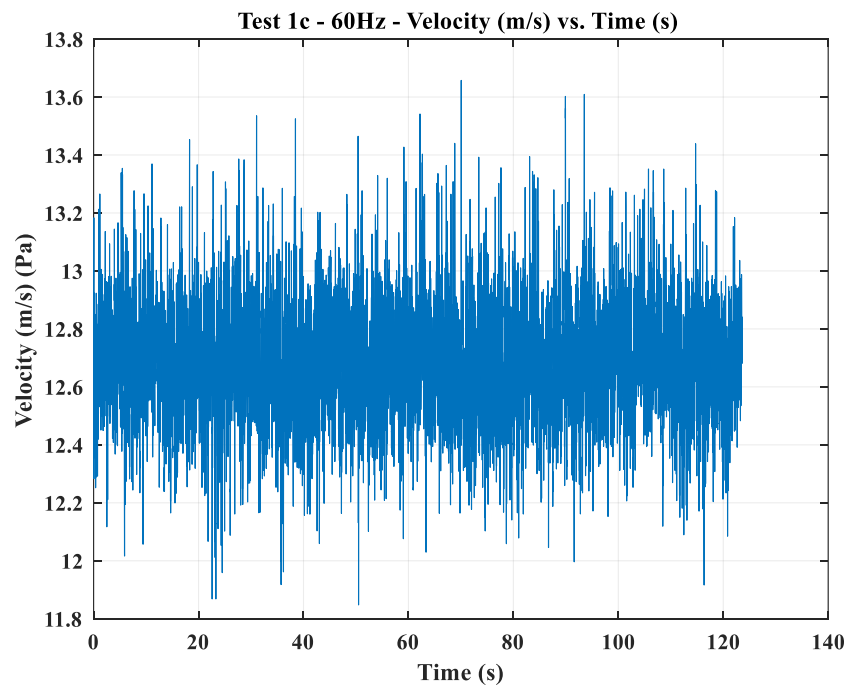


Figure 14: Test 1c (60Hz) Velocity vs. Time Response

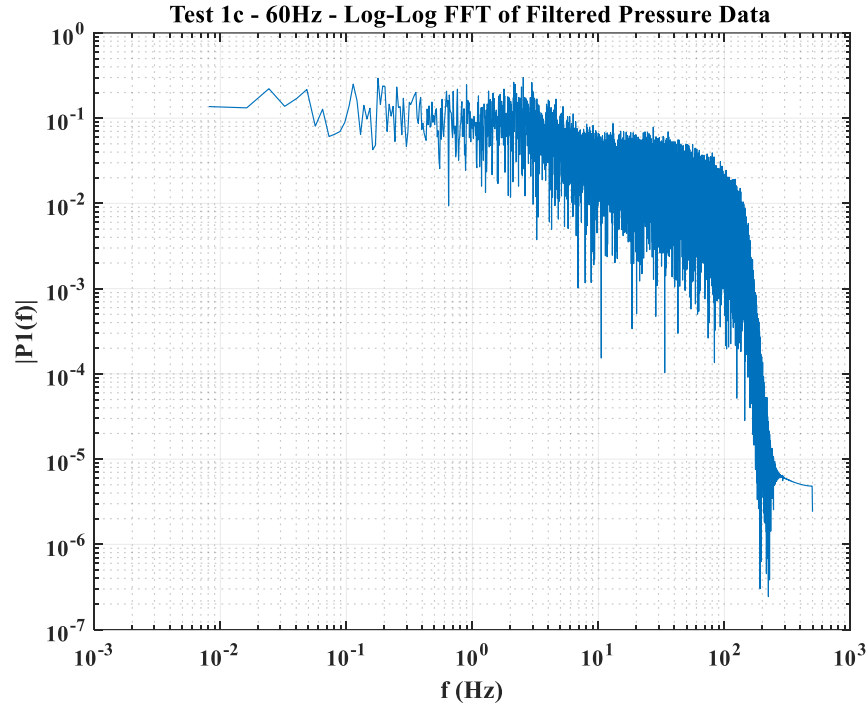


Figure 15: Test 1c (20Hz) FFT Plot

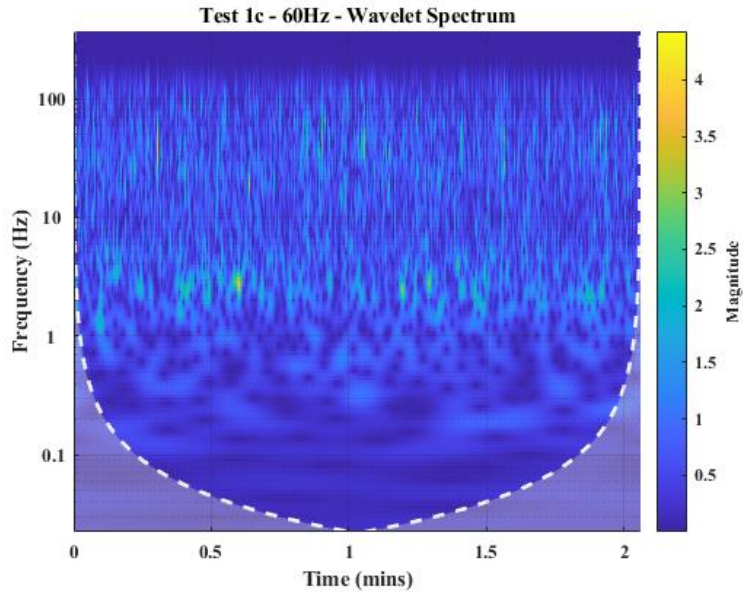


Figure 16: Test 1c (60Hz) Wavelet Spectrum

The final test at 60Hz produced noticeably higher variance in results, however, like the 40Hz fan speed test, velocity readings match closely with expected wind speeds within the tunnel. Notably, based on the wavelet spectrum, a spike at the 0.6-minute mark is observed between the 1 and 10Hz frequency range. This may have caused a disturbance in average velocity readings, where even though they match closely with expected values, they deviate greater than the 40Hz sample.

TI vs. Wind Tunnel Fan Speed Results

A summary of the velocity profiles at each wind speed setting are visible below.

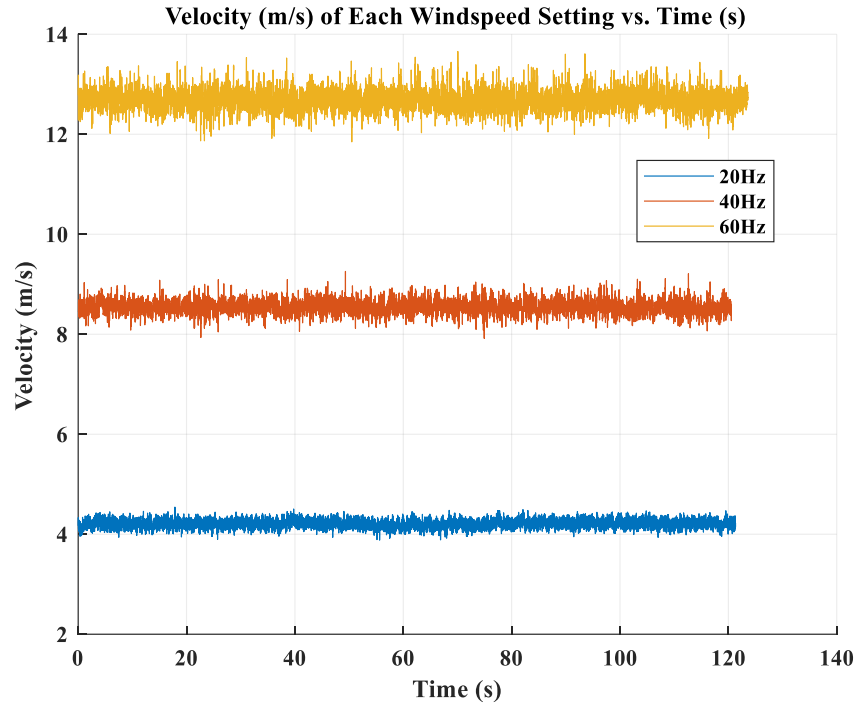


Figure 17: Velocity vs. Time for Each Windspeed Setting

As previously discussed, as the variance of each test increases steadily through each incremental increase in wind speed setting. Since velocity uncertainty is inversely proportional to its free stream velocity, it is expected that higher velocity readings should contain lower variance in data. However, variance at higher velocities represent that fluctuations are a result of the turbulent nature of the flow rather than uncertainties dependent on properties of sensor measurements. This is further characterised by noting a general increase in accuracy of the mean air speed in the free stream increasingly matching expected speeds as the fan speed was increased. Quantitatively, this can also be expressed as a % difference, where a large jump down in error is noticed between the 20hz and 40Hz test samples as displayed in Table (1).

VFD Controller Frequency	Approximate Outlet Speed (m/s)	Mean Outlet Speed (m/s)	% Difference
20Hz	5.5	4.210	23.45
40Hz	8.5	8.546	0.54
60Hz	12.5	12.686	1.48

Table 1: % Error Between Theoretical vs. Actual Free Stream Air Velocity

This relation can be expressed graphically, where the error bars for each sample decrease as the fan frequency is increased. Error is calculated using results from Equation (4) in Part 1 of analysis.

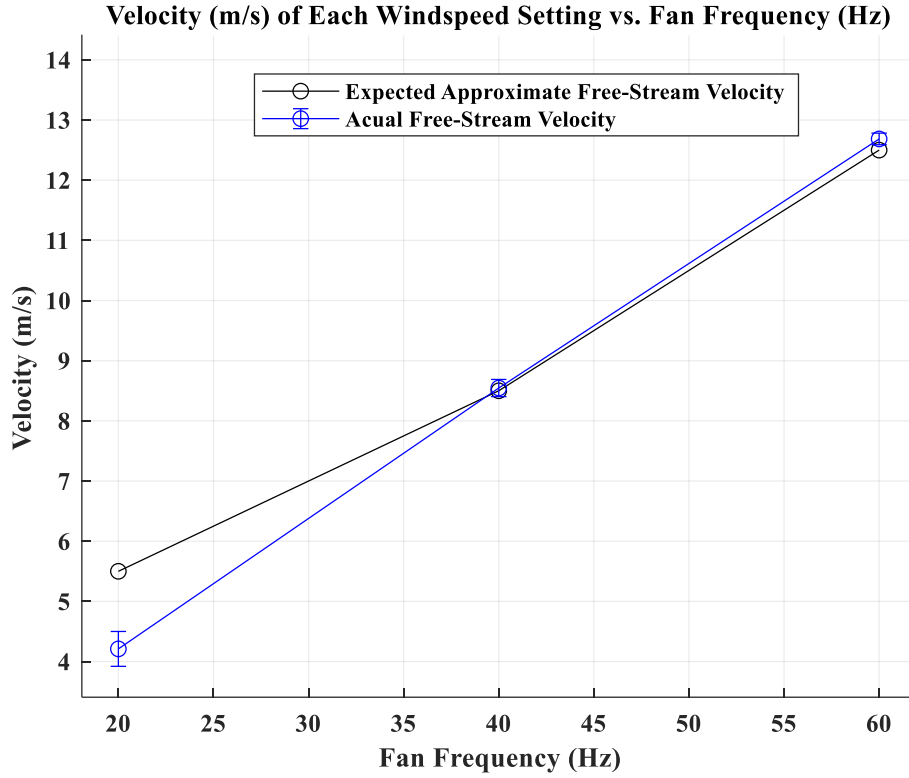


Figure 18: Expected vs. Actual Wind Speeds

Using the standard deviation and mean of velocity readings in each fan frequency test, we can determine the turbulence intensity (%) as a function of fan speed. A sample calculation for both turbulence intensity and the turbulence intensity at 20Hz fan speed.

Assuming stationary flow (meaning there are no fluctuations of the overall variance of data over the sample period), the root mean square of velocity fluctuations is equivalent to the standard deviation of experimental data.

$$TI = \frac{\overline{u_{rms}}}{\bar{V}} = \frac{\sigma_x}{\bar{V}} \quad (6)$$

$$TI = \frac{\sigma_x}{\bar{V}} \cdot 100\% = \frac{0.0677 \frac{m}{s}}{4.2104 \frac{m}{s}} \cdot 100\% = 1.61\%$$

Now, we must calculate the uncertainty corresponding to the 20Hz fan setting turbulence intensity reading. Note that we can neglect uncertainty of the sample standard deviation since the sample size of readings is very large.

$$\delta TI = TI \sqrt{\left(\frac{\delta V}{V}\right)^2 + \left(\frac{\delta \sigma_x}{\sigma_x}\right)^2} \approx TI \left(\frac{\delta V}{V}\right) \quad (7)$$

$$\delta TI = TI \left(\frac{\delta V}{V}\right) = 1.61 \left(\frac{1.22}{4.2104}\right) = 1.61 \left(\frac{1.22}{(4.2104)^2}\right) = \pm 0.1106\%$$

Therefore, the full data point can be expressed as shown below:

$$TI \pm \delta TI = 1.61\% \pm 0.1106\%$$

The full plot for each fan speed is displayed below.

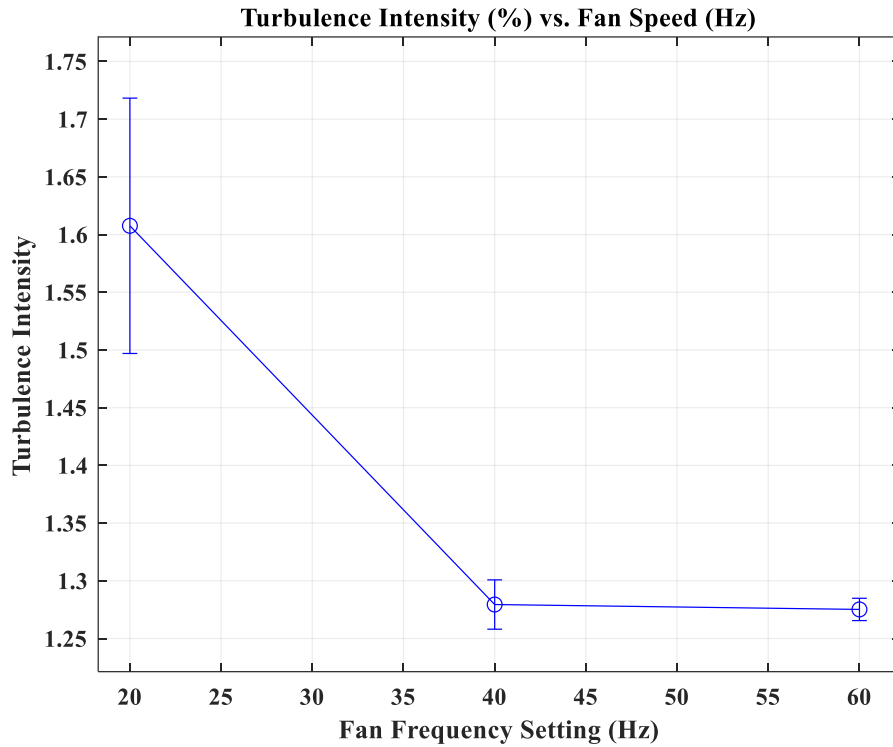


Figure 19: Turbulence Intensity vs. Fan Frequency Setting (Hz)

As a result of direct inverse proportionality of both the uncertainty and turbulence intensity to wind tunnel air velocity, the error band appears much larger for smaller velocities, decreasing in sizes for each sample of increasing fan speed. The turbulence intensity and error band also appears to decrease significantly at 40Hz. This result makes physical sense because as pressure readings in the fan decrease, the sensor noise and hysteresis + sensitivity errors based on sensor specifications become much more dominant in the resulting sensor signal. Since noise becomes dominant, it becomes harder to characterize fluctuations that result from the turbulent nature of the flow itself. This, the turbulence intensity and resulting error bars becomes artificially inflated.

Part III: Part B Results – TI vs. Z-Position Data Collection

As completed for tests in Part A, pressure readings are converted to velocity in all test cases utilizing Equation (5). To effectively determine a distribution of velocity and turbulence intensity across the z-direction of the wind tunnel, tests were measured at points from the top of the wind tunnel to the bottom following the supplemental diagram below. Additionally, both the outer dimensions of the wind tunnel and corresponding dimensions of testing locations are provided on the drawing.

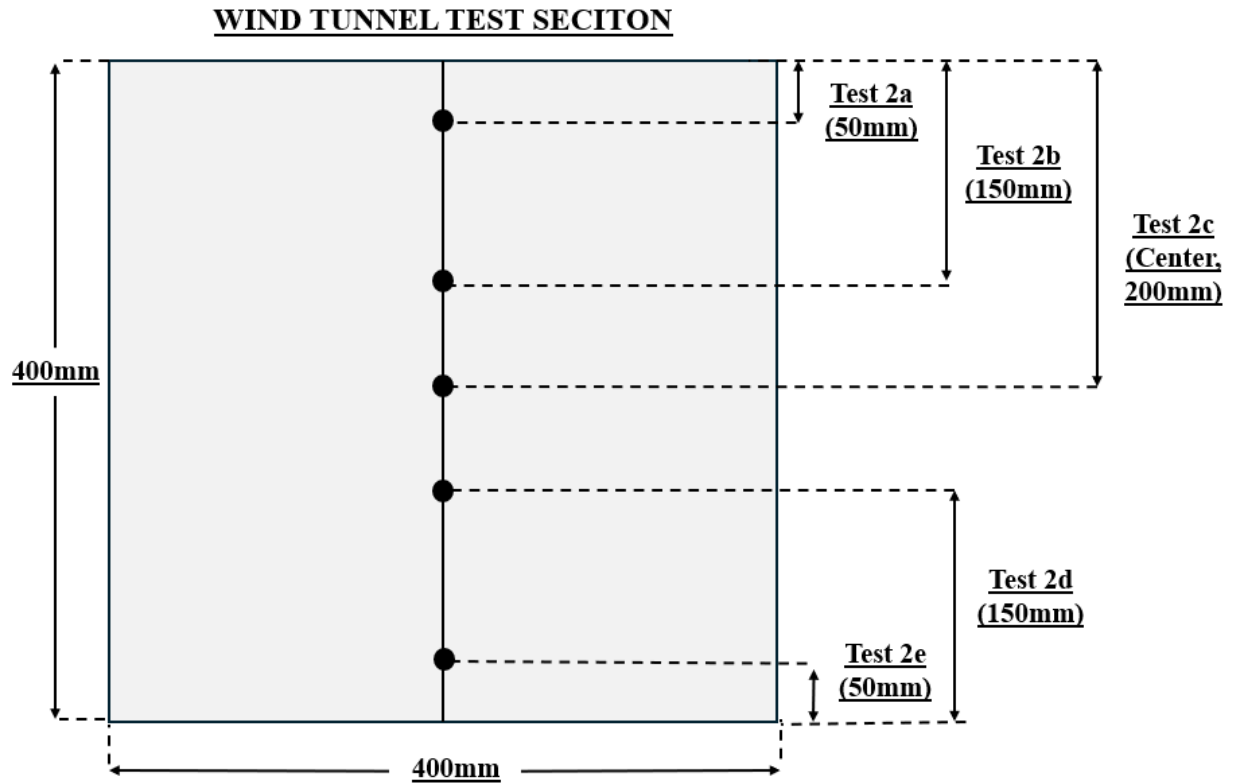


Figure 20: Wind-Tunnel Cross Section Diagram

Test 2a - 50mm From Top of Wind Tunnel Test Results

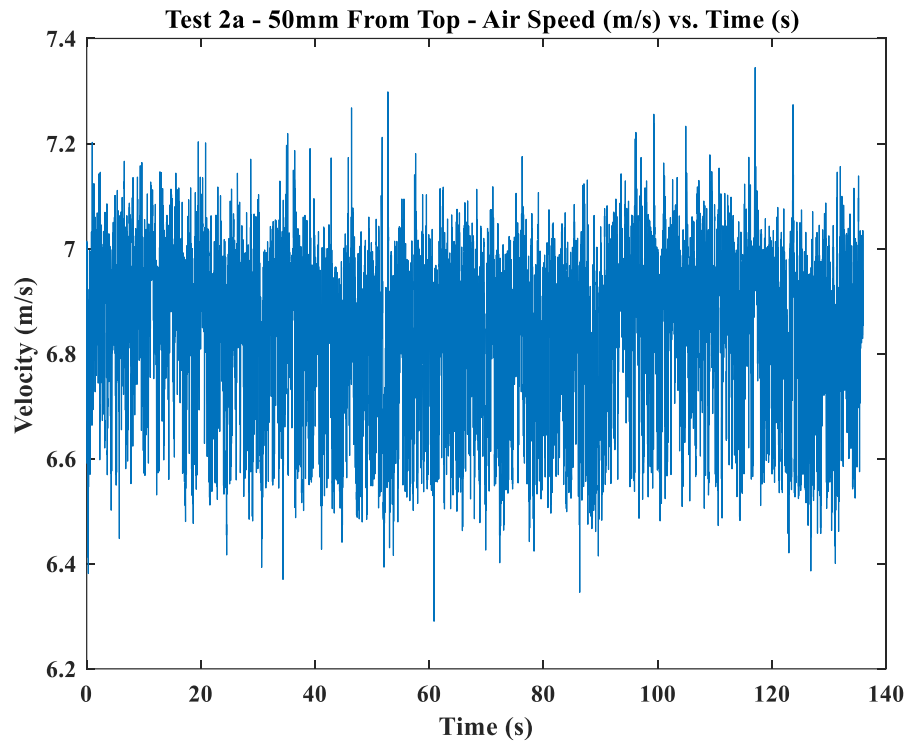


Figure 21: Test 2a (30Hz) Velocity vs. Time Response

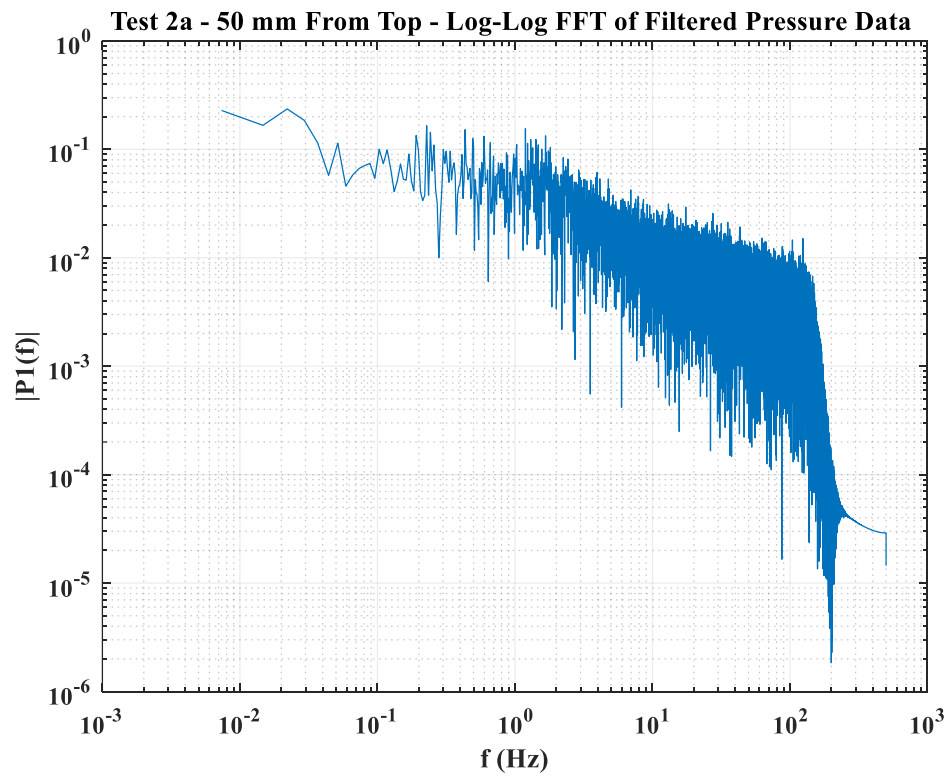


Figure 22: Test 2a FFT Plot

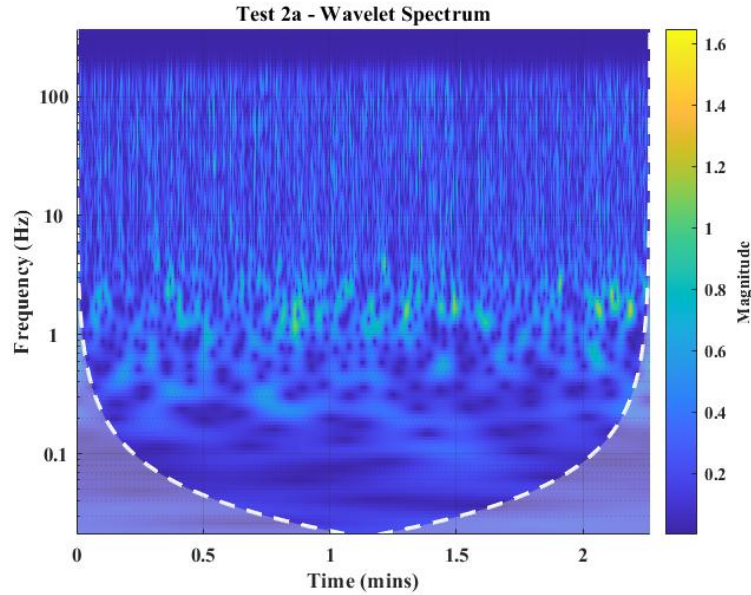


Figure 23: Test 2a Wavelet Spectrum

As displayed based on results from both velocity vs. time and the wavelet spectrum, the test samples collected at 50mm from the top of the air tunnel, large magnitudes at the ~1-5Hz frequency range that are maintained through testing. As such, these frequency spikes will affect the sample variance and cause sufficient on the turbulence intensity at this altitude in the z-direction. These frequency spikes should not be filtered, as they are characteristic of flow separation because of proximity to the boundary of the wind tunnel added with separation from release from the diffuser.

Test 2b - 150mm From Top of Wind Tunnel Test Results

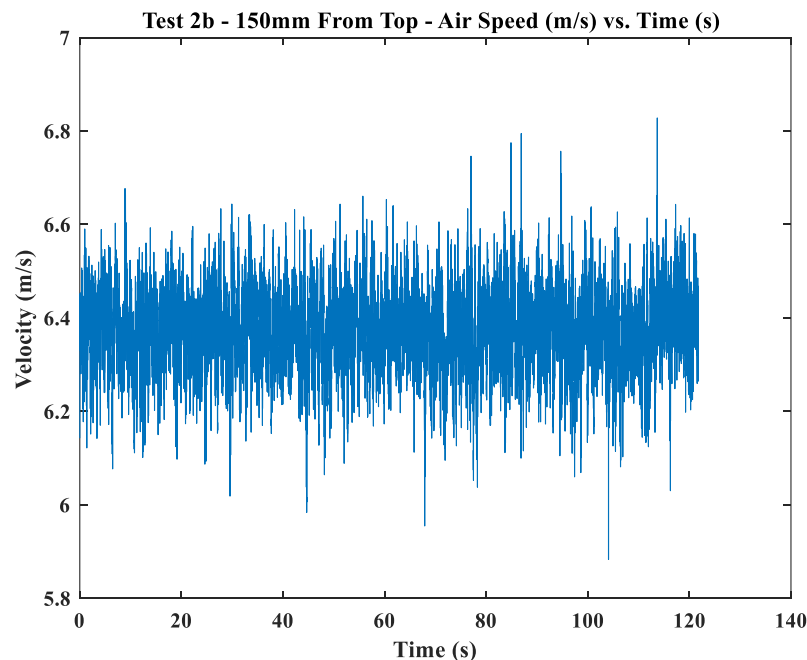


Figure 24: Test 2b (30Hz) Velocity vs. Time Response

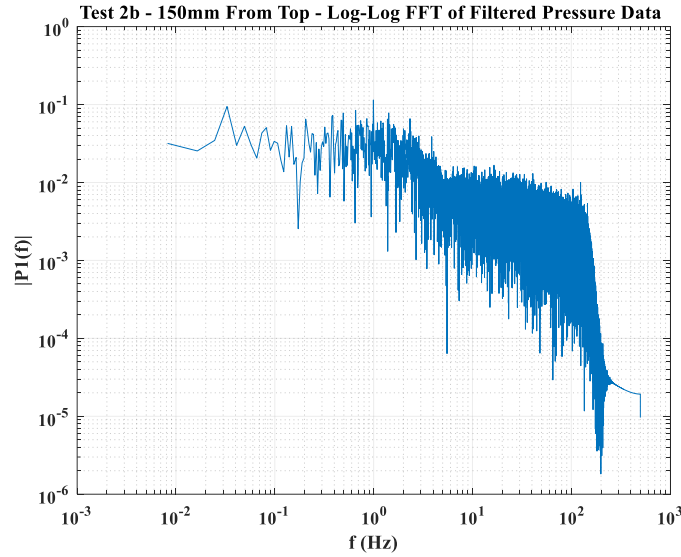


Figure 25: Test 2b FFT Plot

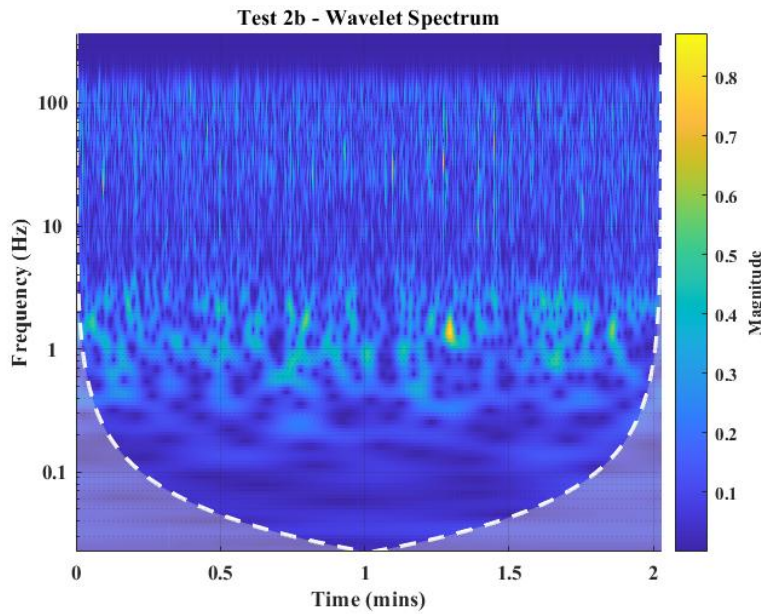


Figure 26: Test 2b Wavelet Spectrum

As visible from testing at 150mm from the top of the wind tunnel, as flow reaches closer to the center of the wind tunnel (Test 2c, 200mm from the top of the wind tunnel), the magnitudes experienced at low frequencies begin to reduce. Aside from a spike at approximately 1.3 minutes into testing at the 1-5Hz frequency range, on average, the magnitudes at the low frequency range are higher in Figure (23) than Figure (26). It is also clearly visible when comparing the velocity readings from the 50mm to 150mm case that there is a noticeable decrease in variance in sensor data. This indicates a trend that velocity closer to the edge of the test section experiences both a larger velocity and an increased turbulence intensity. Since both tests are operated at the same operating fan speed, affects due to precision and bias error from the sensor will be limited compared to affects from testing in Part 1. Essentially, fluctuations in velocity between tests will be less impactful than characteristic variance induced by turbulence in the flow.

Test 2c - Center of Wind Tunnel Test Results

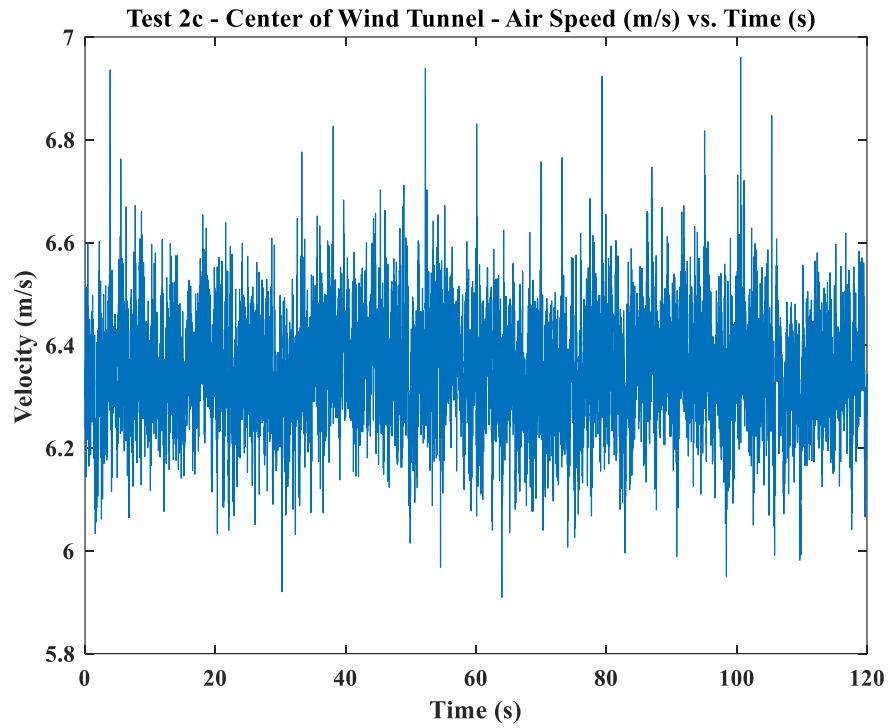


Figure 27: Test 2c (30Hz) Velocity vs. Time Response

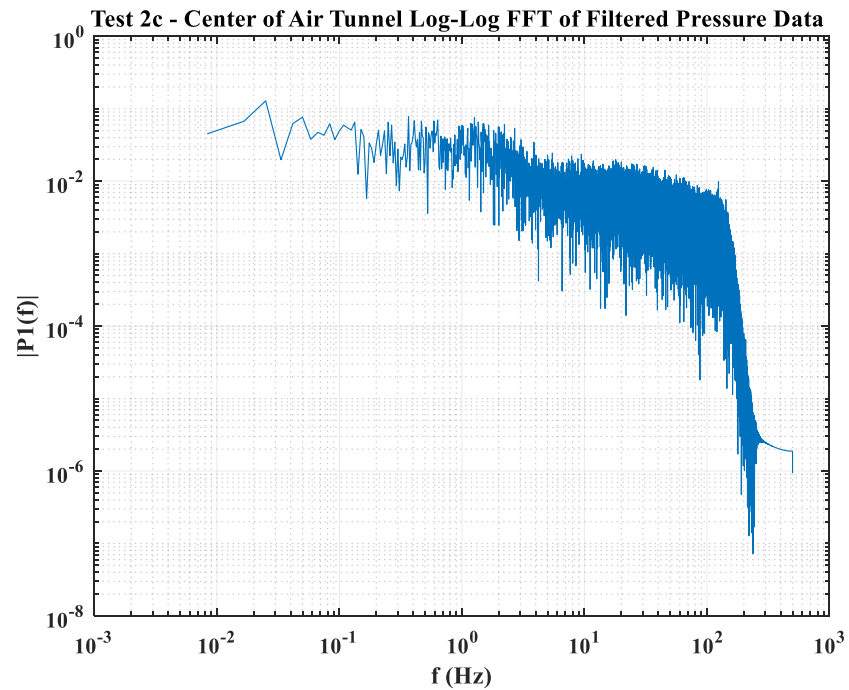


Figure 28: Test 2c FFT Plot

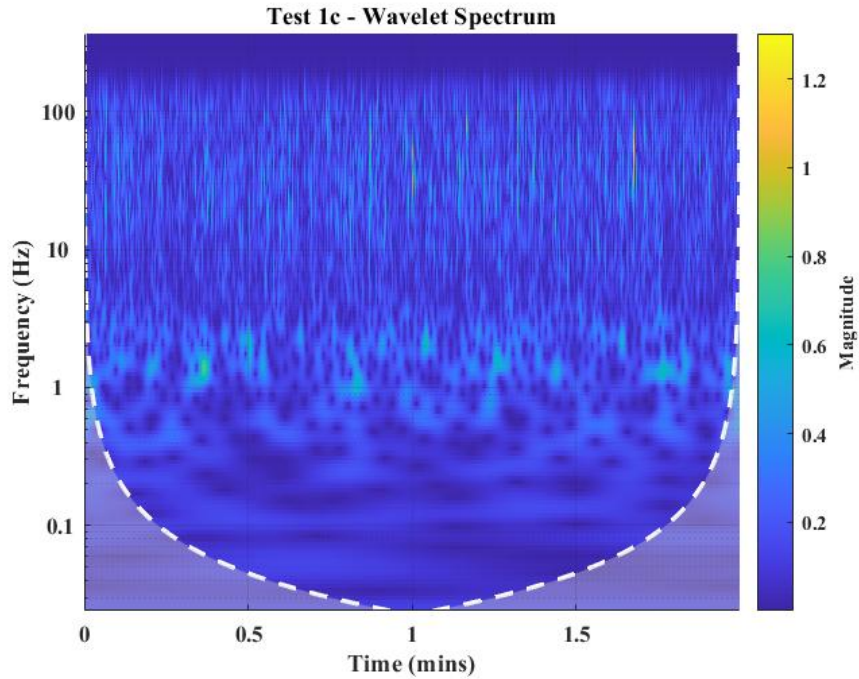


Figure 29: Test 2c Wavelet Spectrum

As expected, at the center of the wind tunnel, velocity is found to be lower than both previous cases (Tests 2a & 2b). However, there is still visible high magnitudes in the 1-5Hz range based on the wavelet spectrum for this case, indicating variance in data. This indicates, therefore, that we should expect the turbulence intensity should be similar, if not slightly larger than the sample case at

Test 2d - 150mm From Bottom of Wind Tunnel Results

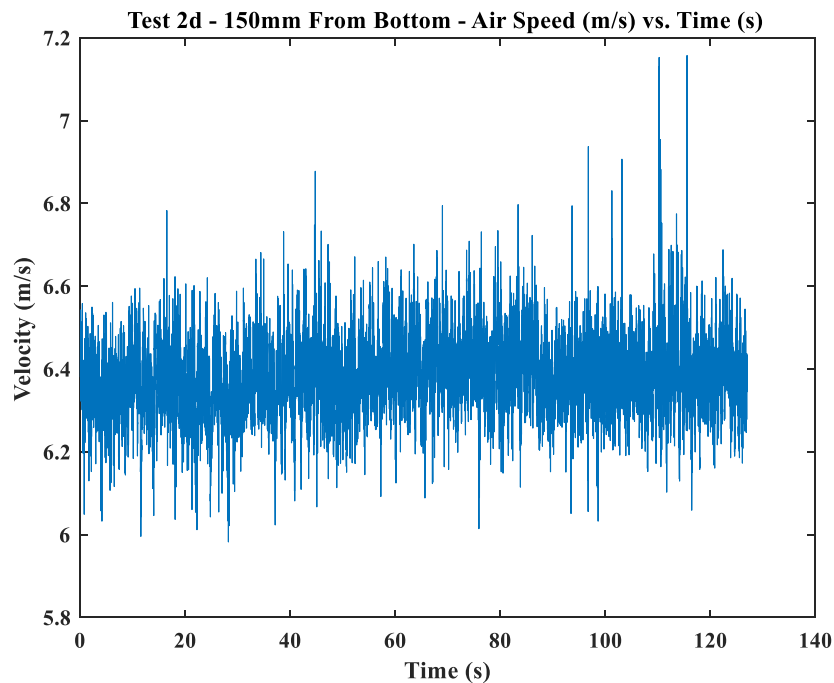


Figure 30: Test 2d (30Hz) Velocity vs. Time Response

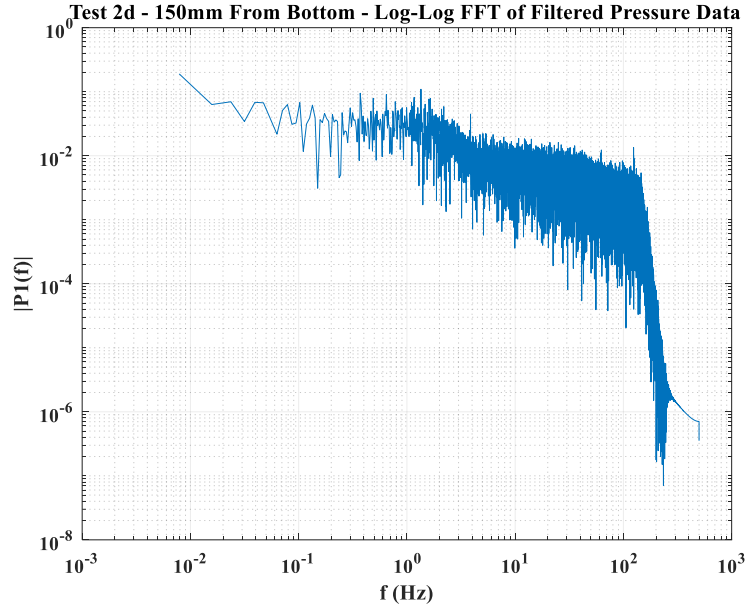


Figure 31: Test 2d FFT Plot

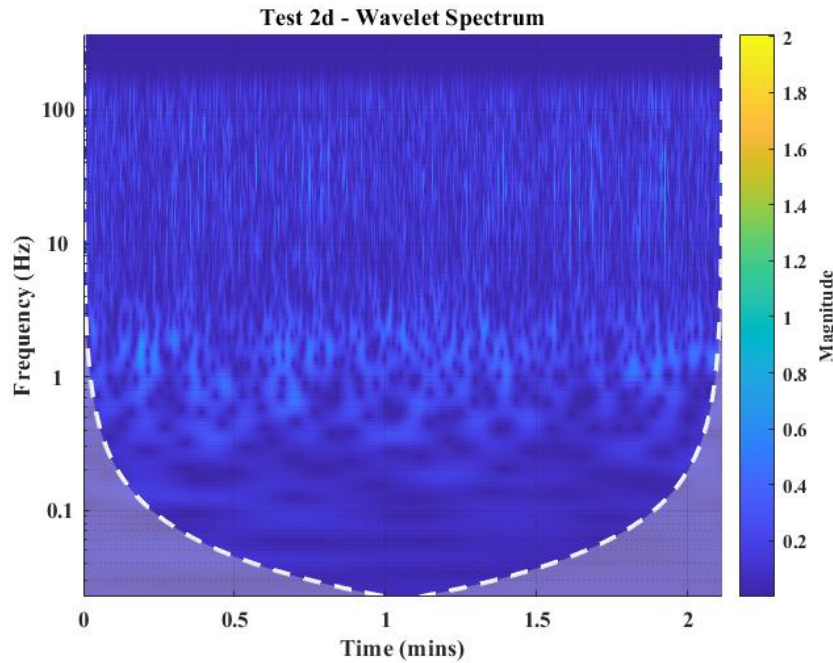


Figure 32: Test 2d Wavelet Spectrum

Based on both the plots of velocity vs. time and the wavelet spectrum plot for Test 2d, we can observe that the mean velocity in this case matches closely with the mean velocity of sample data from test 2b. This validates the existence of a rough symmetrical velocity profile within the wind tunnel. However, it is also worth noting that comparing both wavelet spectrums, there is significantly less noise as visible both Figures (30) – (32). Turbulence looks to have less an impact due to potential affects from the test channel being close to the floor of the experiment environment.

Test 2e - 50mm From Bottom of Wind Tunnel Results

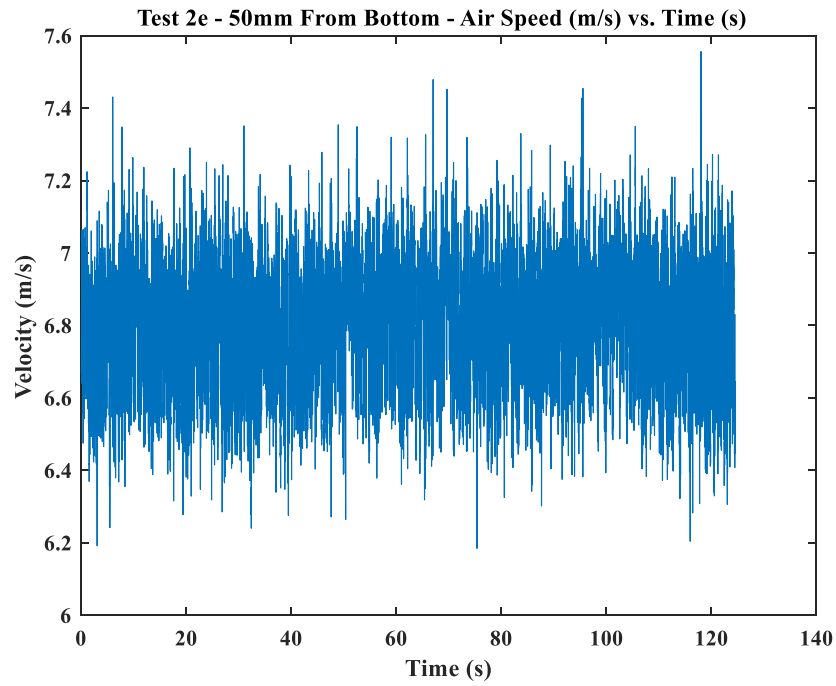


Figure 33: Test 2e (30Hz) Velocity vs. Time Response

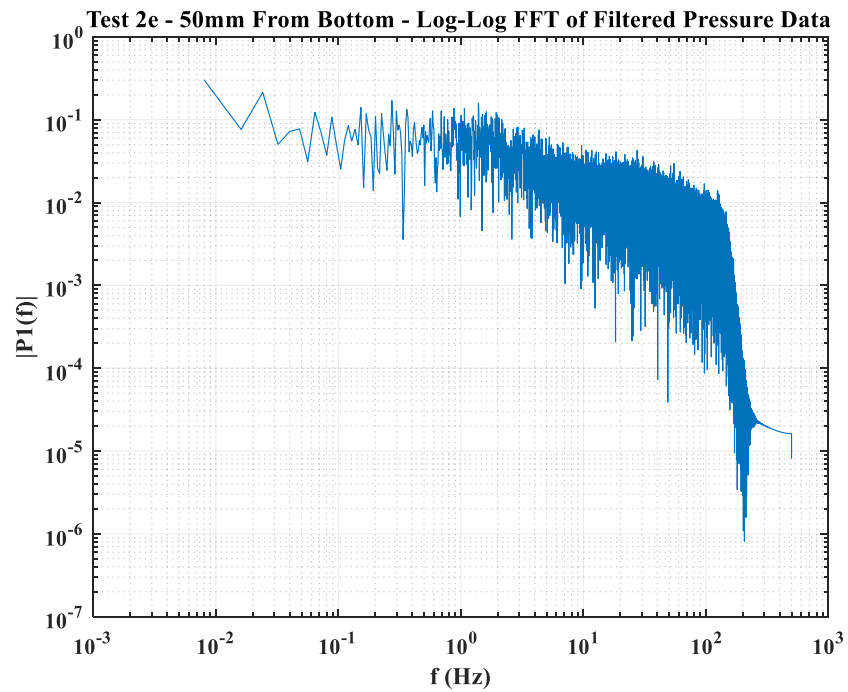


Figure 34: Test 2e FFT Plot

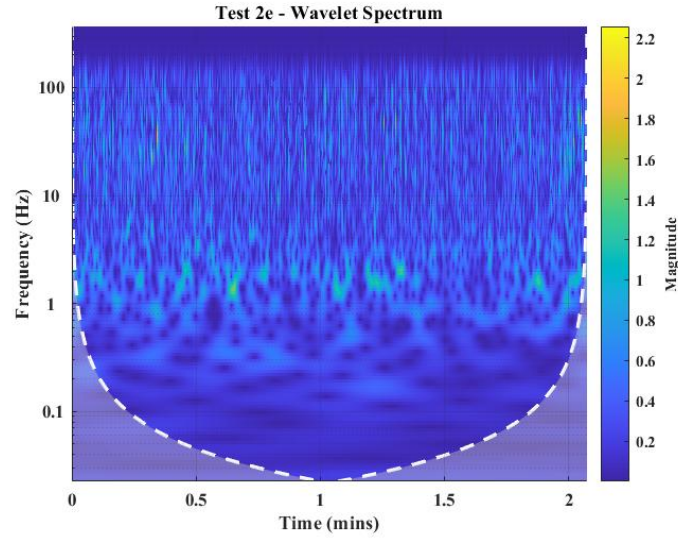


Figure 35: Test 2e Wavelet Spectrum

The final test as apart of Test 2, where data was measured 50mm from the bottom of the wind tunnel indicates and validates a symmetric impact of both turbulence intensity and the overall velocity profile. The velocity profile from Figure (33) shows visibly higher noise at a higher velocity. The noise would dominate when determining the turbulence intensity, this producing similar expected results to test 2a. No large frequency spots can be in Figure (35), the wavelet spectrum, for Test 2e.

Velocity & TI vs. Z-Position Data Collection Results

As an initial comparison, all plots of velocity vs. time for each case in Test 2 are displayed on the same graph.

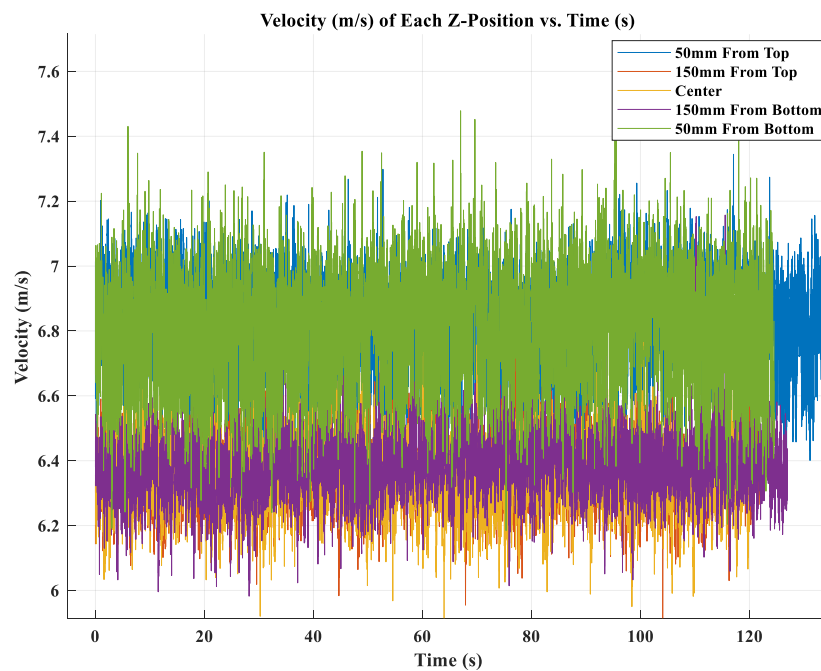


Figure 36: Velocity (m/s) vs. Time (s) for All Z-Direction Samples

As expected from what was observed from results for each individual test case, Tests where the pitot tube was positioned closer to the edge of the wind tunnel experienced a higher overall velocity coupled with an increased sample variance. To better visualize this data, the mean velocity is computed for each case plotted against the relative position along the z-direction of the wind tunnel to form an estimate of the velocity profile in the wind tunnel. An error bar is provided based on total uncertainty of velocity readings found through Part 1. Notably, due to small fluctuations in velocity, the error bar will remain similar for all data points. For visual aid, the positive positions would correspond to the distance from the center of the wind tunnel upwards, and the negative positions in the opposite, downward direction.

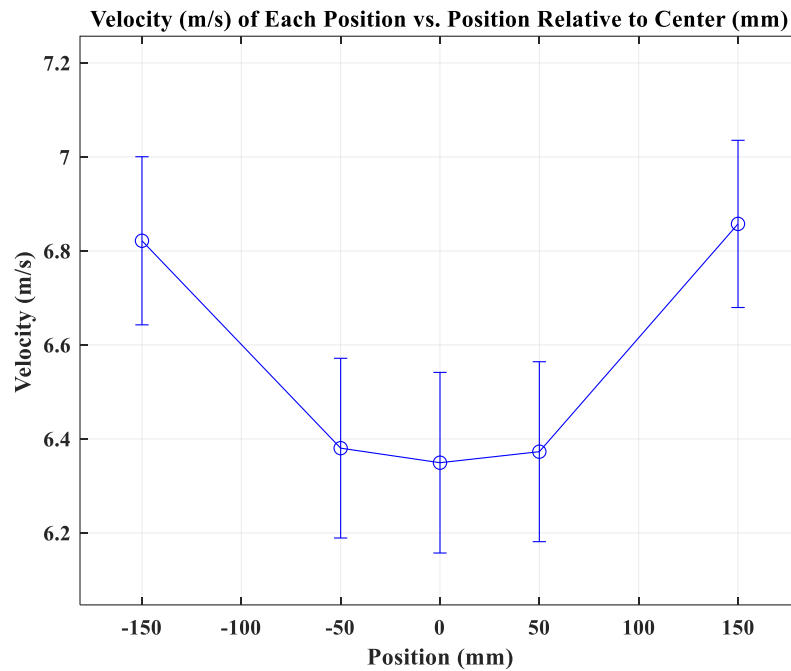


Figure 37: Velocity Profile of Wind Tunnel

As displayed in Figure (7), expected symmetry within the wind tunnel is observed. Given this distribution, along with relatively symmetric sample variances across the wind tunnel section, we should expect that the turbulence intensity should also be approximately symmetric across the profile of the wind tunnel. Given the nature of fluid motion within a confined cross section, where a convex velocity profile is expected to be achieved, this data produces an inverse profile to what is expected. In general, there are two potential causes of this distribution, namely debris and clogging within the honeycomb flow conditioner located near the inlet of the test section; and one source of error being low-velocity wakes that persist at the center of flow due to the small size of the wind tunnel. These can be resolved by (1) regularly maintaining the conditioner to reduce debris, and (2) redesigning the wind tunnel such that flow has more length to form a uniform velocity profile.

Similarly to Test A, the turbulence intensity is to be evaluated for each sample. A sample calculation for Test 1a (50mm from the top of the tunnel). This is also shown visually at the 150mm position on Figure (37).

$$TI = \frac{\sigma_x}{\bar{V}} \cdot 100\% = \frac{0.1312 \frac{m}{s}}{6.8575 \frac{m}{s}} \cdot 100\% = 1.91\%$$

$$\delta TI = TI \left(\frac{\delta V}{V} \right) = 1.91 \left(\frac{1.22}{V^2} \right) = 1.91 \left(\frac{1.22}{(6.8575)^2} \right) = \pm 0.0497\%$$

Therefore, the turbulence intensity at Position = 150mm (50mm from the top of the air tunnel) is:

$$TI \pm \delta TI = 1.91\% \pm 0.0497\%$$

The full plot, along with respective uncertainties are displayed below.

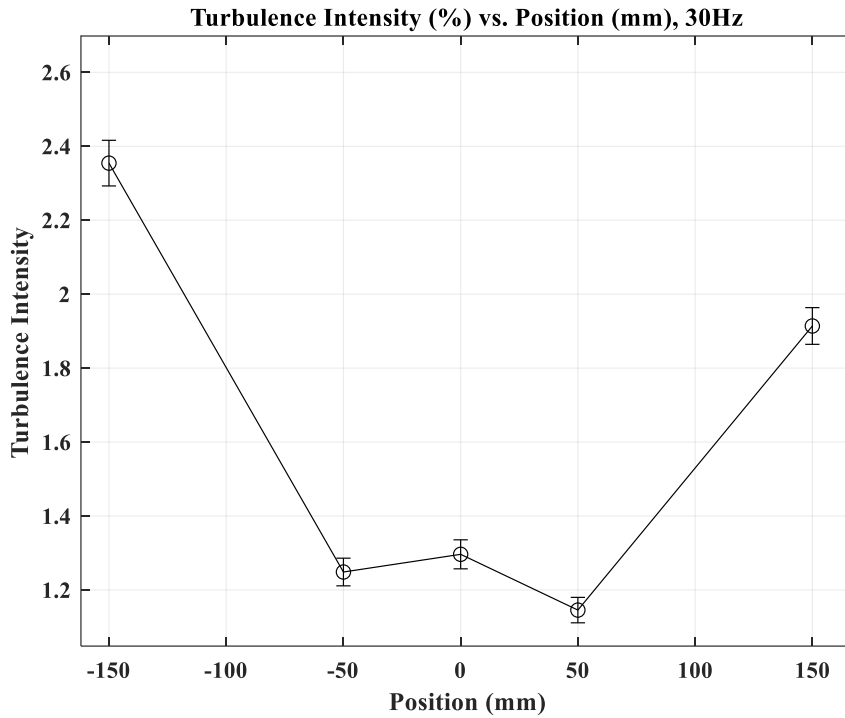
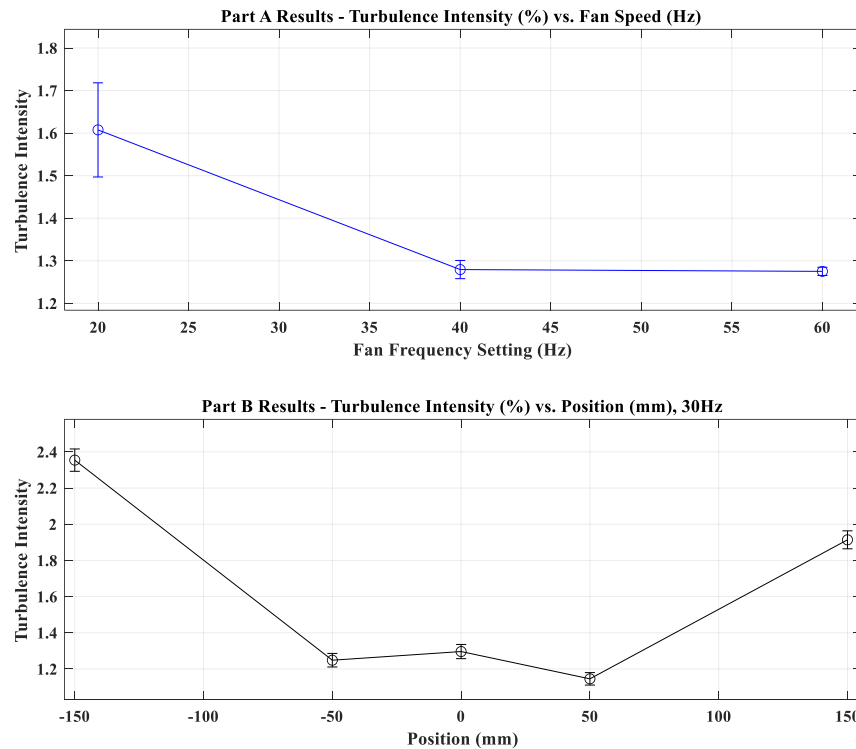


Figure 38: Turbulence Intensity (%) vs. Position (mm)

As expected, the turbulence intensity, being proportional to variance of velocity data, experiences it's highest magnitude when the flow is closer to the edges of the cross section/wind tunnel, decreasing in variance as the flow approaches the center of the wind tunnel cross section. Error bars are also maintained at a relatively small margin of error due to pressure readings recorded significantly above the noise baseline, preventing artificial inflation of expected error. In Part IV, discussion will be placed on comparing turbulence intensity results between fan speeds and z-direction positioning to determine final conclusions and implications of results.

Part IV: Summary Results & Conclusion

The results from Parts II and III (Part A - Tests 1abc, Part B – Tests 2abcde).



Based on findings from both Parts A & B, there are two major characteristics of flow with high turbulence intensity. First, when the flow is placed closer to the edge of the wind tunnel, turbulence increases, as seen with a significantly higher variance in data, the turbulence intensity significantly increases to nearly double then flow at the center of the wind tunnel. Secondly, turbulence intensity is also directly proportional to a decrease in operating wind speed. Although this jump is not as large as a change in position along the z-axis, low velocity readings, indicating low pressure readings, become hard to read due to interferences of noise signals in data processing. The velocity reading at the 20Hz sample was also artificially decreased due to non-linearity of the fan near the limits of its operating range. Resultingly, the turbulence intensity at 20Hz should be lower than it is. This highlights not just the impact of wind speed on turbulence intensity, but data analysis errors that result from selecting inadequate sensing equipment for given ranges of expected differential pressure, as well as selecting adequate fan speed settings that function within intended operating limits to assess fluid characteristics.

In conclusion, along with characterising turbulent intensity within the Steady wind tunnel, this assignment highlighted the importance of sensor selection, the impacts of appropriate filtering, and how error propagation and uncertainty is affected by the testing environment and validated within approved calibrated output ranges. Ensuring that an appropriate sensor range is chosen is critical to ensure that desired output values are far outside the range of baseline noise, and appropriate shakedown tests are critical to ensure that data collection methodology and analysis methods are sound to make reasonable conclusions on fluid nature rather than artificial impacts.